Structural Balance in the Bitcoin Alpha Trust Network

Positive and negative trust relationships in a signed social network

85.64%

of observed closed triads are structurally balanced

Compared with 76.94% under a fixed-topology sign shuffle

Submitted by: _____ USN: _____

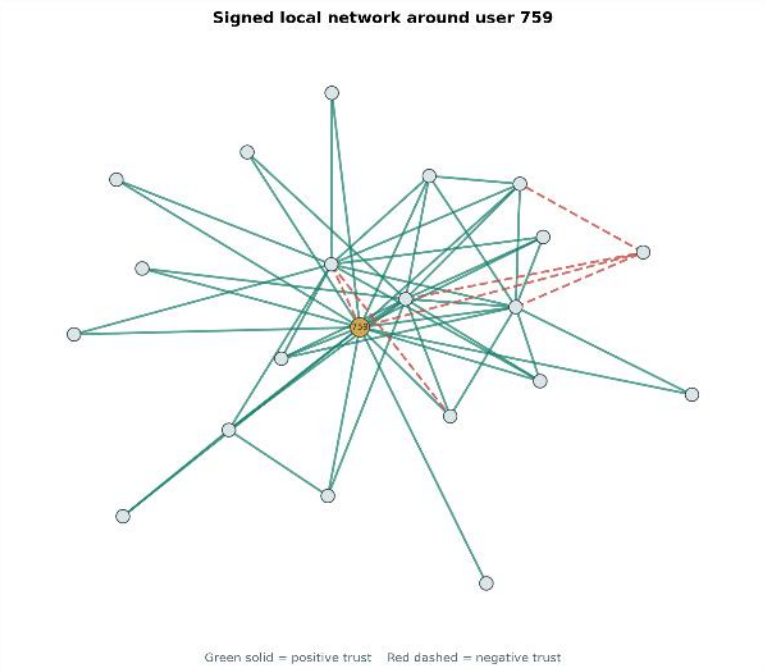
Submitted to: Course Instructor | Social Network Analysis

Users rate trust from -10 to +10

3,783
user nodes

24,186
directed ratings

Each rating is a signed relationship from the rater to the ratee. Positive values express trust; negative values express distrust.



One local neighborhood. Green lines are positive, red dashed lines are negative.

Structural balance needs an undirected signed graph

To evaluate balance, reciprocal ratings are averaged for each unordered user pair.

01

24,186

directed ratings

02

14,124

unordered pairs

03

14,081

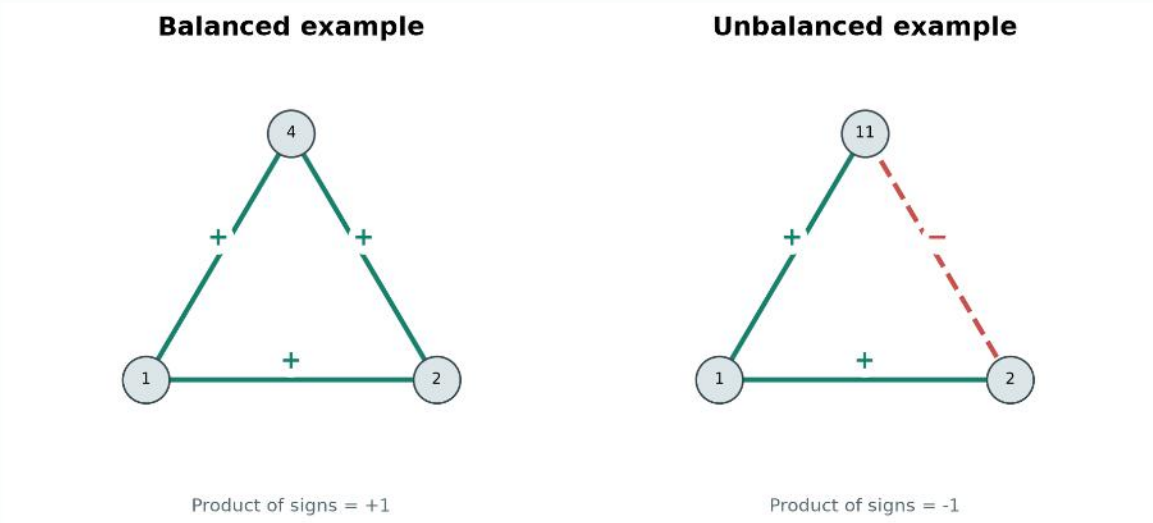
non-neutral signed ties

43 zero-mean pairs are excluded as neutral. The projection can hide disagreement between reciprocal ratings.

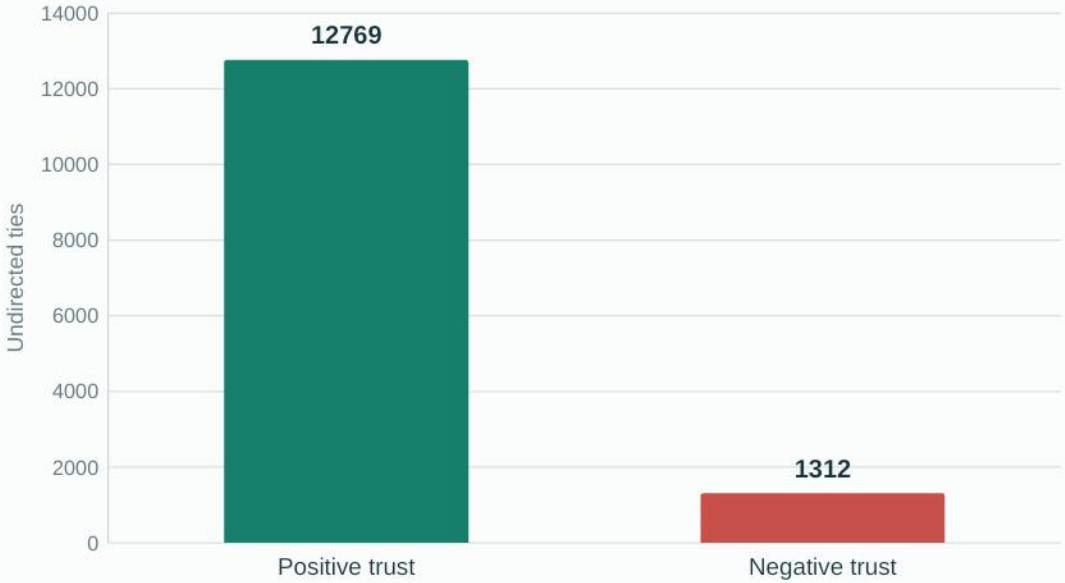
12,769 positive ties **1,312 negative ties**

A triad is balanced when its sign product is positive

For each closed triad, multiply the three edge signs.



Positive trust dominates the projected ties



90.68%

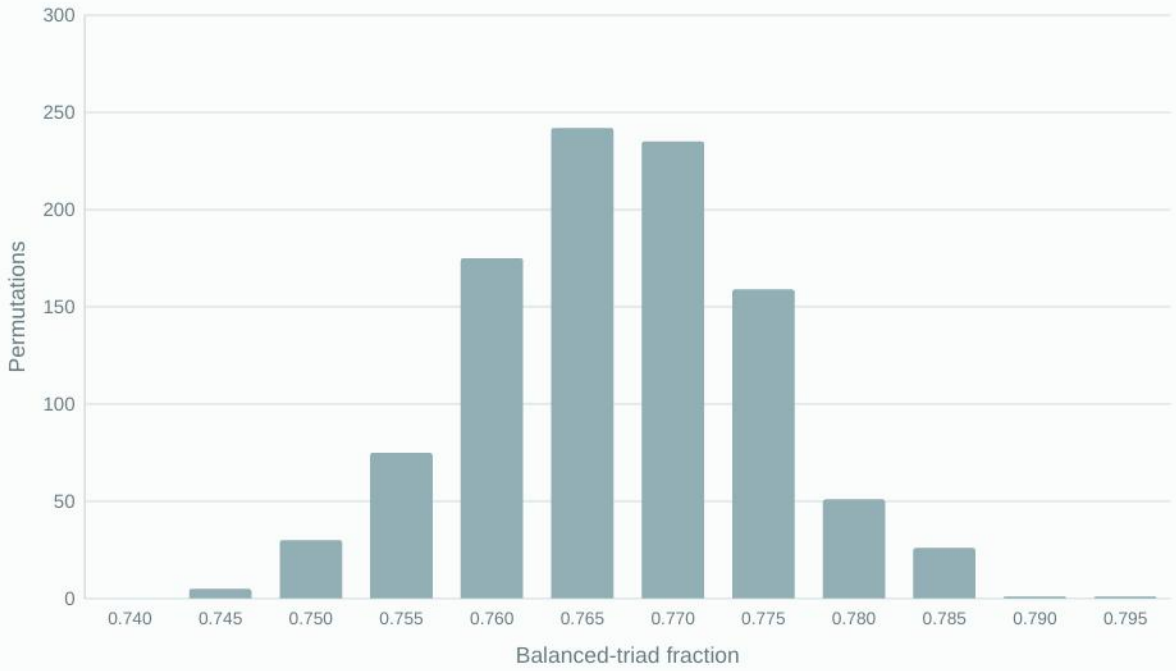
of non-neutral ties are positive

14,081

signed ties in the projection

The graph is sparse: density 0.00197. Its largest component contains 99.71% of users.

Observed balance exceeds the sign-shuffle range



OBSERVED

85.64%

Null mean

76.94%

95% null interval

75.36%-78.52%

1,000 sign permutations
Plus-one p = 0.001

The result describes local triads, not every relationship

18,565
balanced triads

3,112
unbalanced triads

The observed balance rate is 8.70 percentage points above the null mean. Under this null, signs are arranged into more balanced closed triads than random reassignment would produce.

> **Interpret with care**
>
> Averaging reciprocal ratings can hide disagreement. The null keeps overall sign counts, but not each user's sign pattern. The data are historical and do not describe current platform behavior.

Conclusion and study scope

Conclusion

Bitcoin Alpha has 85.64% balanced closed triads, above the 76.94% mean under the fixed-topology sign shuffle.

Interpretation boundary

The 85.64% result describes closed triads in an undirected projection. The null preserves graph topology and total sign counts, but not each user's sign tendencies. The analysis does not establish causality or a global two-faction structure.

Data and citation

Kumar et al. (2016), Edge Weight Prediction in Weighted Signed Networks.
SNAP dataset: snap.stanford.edu/data/soc-sign-bitcoinalpha.html